Teacher: Mr Munandi ZP
School: Tswinga Primary School
Grade: 7
Date: 23 April 2026
Duration: 30 minutes

CAPS Topic:

Numbers, Operations and Relationships
(Sub-topic: **Exponents**)

Specific Aims (CAPS):

Learners should be able to:

- Develop understanding of numbers and their relationships.
- Use appropriate techniques to perform calculations.
- Solve problems involving numerical expressions.

Lesson Objectives:

By the end of the lesson, learners should be able to:

1. Recognise and describe exponents as repeated multiplication.
2. Apply the laws of exponents for whole numbers.
3. Simplify numerical expressions involving exponents.

Resources (LTSM):

- Chalkboard and chalk/marker
- Prepared number/exponent cards
- CAPS-aligned textbook
- Worksheet/exercises

Introduction (5 minutes):

- Revise prior knowledge: multiplication as repeated addition.
- Ask:
  - "How can we write $2 \times 2 \times 2$ in a shorter way?"

- Introduce exponents as repeated multiplication.
- Examples:
  - $2^3 = 8$
  - $3^2 = 9$
- Explain terminology: **base** and **exponent (index)**.

Lesson Development (20 minutes):

Activity 1: Concept of Exponents (5 minutes)

- Learners expand expressions:
  - 4^2, 5^3
- Teacher emphasises correct notation and meaning.

Activity 2: Laws of Exponents (10 minutes)

Teach CAPS-required exponent laws (for whole numbers):

1. **Product Law:**
   - am×an=am+na^m × a^n = a^{m+n}am×an=am+n
   - Example: $2^2 \times 2^3 = 2^5$
2. **Quotient Law:**
   - am÷an=am−na^m ÷ a^n = a^{m-n}am÷an=am−n
   - Example: $2^5 \div 2^2 = 2^3$
3. **Power Law:**
   - (am)n=amn(a^m)^n = a^{mn}(am)n=amn
   - Example: $(2^3)^2 = 2^6$
4. **Zero Exponent:**
   - a0=1a^0 = 1a0=1, where a≠0a ≠ 0a□=0

- Work through examples with learners.
- Emphasise **same base requirement**.

Activity 3: Application (5 minutes)

- Learners simplify:
  - $3^2 \times 3^3$
  - $4^6 \div 4^2$
  - $(2^2)^3$
- Teacher provides support and correction.

Conclusion (5 minutes):

- Summarise key concepts:
  - Meaning of exponents
  - Laws of exponents
- Oral questions for consolidation:
  - “What happens to exponents when we multiply?”
  - “What is 5^0?”

Assessment (Informal – CAPS aligned):

- Continuous observation during activities.
- Oral questioning.
- Written responses to classwork problems.

Assessment Standard:

- Simplifies numerical expressions involving exponents.

Differentiation:

- **Support:** Provide simpler examples (e.g., $2^2 \times 2^1$).
- **Extension:** Challenge learners with mixed operations.

Homework (CAPS-aligned):

Simplify:

1. $2^3 \times 2^4$
2. $5^6 \div 5^2$
3. $(3^2)^3$